

George Dasoulas

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EDUCATION

APRIL 2019 - Today	École Polytechnique , Palaiseau, Paris, France. • PhD candidate in Computer Science, focused on Machine Learning and Graph Theory. • Laboratory: Data Science and Mining Team, LIX, École Polytechnique. • Funding: CIFRE PhD Collaboration with Huawei Technologies, France. • Thesis title: <i>"The expressive power of graph neural networks and its impact on structured representation learning"</i>. • Thesis supervision: Prof. Michalis Vazirgiannis (DaSciM). • External advisors: Dr. Aladin Virmaux (Huawei), Dr. Kevin Scaman (INRIA Paris). • Planned defense date: March 2022.
OCTOBER 2012 - FEBRUARY 2018	National Technical University of Athens , Athens, Greece. • BSc & MSc in Electrical and Computer Engineering (5-year joint degree; 300 ECTS) – Order of Entrance: 10th among undergraduate students – Overall GPA: 9.06/10 (Excellent). Top 3% among graduate class – Specialization: Computer Science (Major), Mathematics, Networks (Minor) • Thesis : <i>"Predictive Refueling - Combining machine learning algorithms with optimization calculations for operations planning"</i>. Grade: 10/10. • Thesis supervision: Prof. Nectarios Koziris (National Technical University of Athens), Dr. Florian Mittag (TWT GmbH Science & Innovation).
JULY 2012	Nationwide University Entrance Examination score : 19.676/20.000 (top 1% nationwide).

PROFESSIONAL EXPERIENCE

October 2018 - Today	Huawei Technologies , Paris, France • Doctoral researcher in graph machine learning. • Laboratory: Noah's Ark Lab, AI department of Huawei R&D. • Research focus: Representation learning algorithms for static and dynamic graphs. • Project focus: Development of theoretical and practical methods for structured representation learning, specialized in biomedical and communication networks.
June 2017 - January 2018	NIKI Ltd Digital Engineering , Ioannina, Greece & TWT GmbH Science & Innovation , Stuttgart, Germany. • MSc Thesis research in machine learning for prediction-based optimization. • Project: "Predictive Mechanism for optimal refueling strategies" – Benchmarking of ML algorithms for time series forecasting. – Recommender system implementation for short-term prediction of fuel price schemes and decision making on optimal refueling.
October 2013 - February 2016	Software Engineering Laboratory at NTUA, Greece • Laboratory Assistant for computer programming courses. • Tasks: Part-time teaching (twice a week), supporting in laboratory exercises undergraduate students and overseeing the final laboratory examination procedure. • Pertinent Professor: Nikolaos Papaspyrou
September 2014 - January 2017	Private Tutoring , Athens, Greece • Tutor for high-school and undergraduate students • Courses: Linear Algebra, Calculus, Electromagnetism, Computer Programming

PUBLICATIONS

Conferences

1. George Dasoulas, Kevin Scaman, Aladin Virmaux. Lipschitz Normalization for Self-Attention Layers with Application to Graph Neural Networks. In *International Conference on Machine Learning (ICML)*, 2021.
2. Michail Chatzianastasis, George Dasoulas, Georgios Siolas, Michalis Vazirgiannis. Graph-based Neural Architecture Search with Operation Embeddings. In *Neural Architectures: Past, Present and Future Workshop for International Conference on Computer Vision (ICCV)*, 2021.
3. George Dasoulas, Johannes Lutzeyer, Michalis Vazirgiannis. Learning Parametrised Graph Shift Operators. In *International Conference on Learning Representations (ICLR)*, 2021.
4. George Dasoulas, Giannis Nikolentzos, Kevin Scaman, Aladin Virmaux, Michalis Vazirgiannis. Ego-based Entropy Measures for Structural Representations on Graphs. In *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.
5. George Dasoulas, Ludovic Dos Santos, Kevin Scaman, Aladin Virmaux. Coloring graph neural networks for node disambiguation. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2020.
6. Stratis Limnios, George Dasoulas, Dimitrios M. Thilikos, Michalis Vazirgiannis. Hcore-Init: A graph-based initialization method for neural networks. In *International Conference of Pattern Recognition (ICPR)*, 2020.

Journals

7. Giannis Nikolentzos, George Dasoulas, Michalis Vazirgiannis. k-hop Graph Neural Networks. *Neural Networks*, Volume 130, Pages 195-205, October 2020.

Patents

8. George Dasoulas, Ludovic Dos Santos, Kevin Scaman, Aladin Virmaux. Node Disambiguation. WO 2021/058096 A1 International Publication Number. April 2021.

Under Submission

9. Giannis Nikolentzos, George Dasoulas, Michalis Vazirgiannis. Permute Me Softly: Learning Soft Permutations for Graph Representations, 2021.
10. Guillaume Salha-Galvan, Johannes Lutzeyer, George Dasoulas, Romain Hennequin, Michalis Vazirgiannis. Modularity-Aware Graph Autoencoders for Joint Community Detection and Link Prediction, 2021.

HONORS, AWARDS & CERTIFICATES

November 2020	Eleftheria Barka Scholarship. Scholarship for graduation excellence and pursuit of doctoral studies for the academic year 2020-2021.
November 2019	Eleftheria Barka Scholarship. Scholarship for graduation excellence and pursuit of doctoral studies for the academic year 2019-2020.
December 2015	Papakyriakopoulos Award. Award of excellence in Mathematics (corresponding subjects in Mathematics).
March 2014	Silver Medal in South Eastern Mathematical Olympiad for university students (aka. See-mous), Iasi, Romania.
October 2013	Participation in the IEEEExtreme 24-Hour Programming Competition.
September 2012	“The Great Moment of Education” Award – Scholarship by Eurobank EFG, for excellency in national qualifications exams.
May 2010	Certificate for the theoretical and practical educational seminar for C.P.R - F.B.A.O - A.E.D knowledge, Red Cross, Greece.
March 2009	Bronze Medal in National Mathematical Olympiad for high school students in Athens.

COMPUTER SKILLS

Operating Systems: Ubuntu, Debian, Unix, Windows, MacOS.

Programming Languages: C, C++, Python, R, Java, SQL, Haskell, OCaml, Prolog, MATLAB, bash, Latex.

SOCIETIES & AFFILIATIONS

- IEEE Transactions on Neural Networks and Learning Systems, Reviewer, 2020-Today.
- IEEE Computer Society, Member, 2013-2014.

TEST SCORES

- General GRE: Quantitative Score: 167/170, Verbal Score: 157/170, Essay: 4/6.
- TOEFL iBT : 100/120.

LANGUAGES

- **Greek** (Native speaker)
- **English** (Excellent - CEF:C2) : Examination for the Certificate of Proficiency in English (ECPE)
- **German** (Very good command - CEF :C1) : Goethe-Zertificat B1, TestDAF Level C1
- **French** (Intermediate)